

ParkShare – Problem Statement

AI-Powered Private Parking Space Sharing and Smart Parking Recommendation Platform

1. Problem Overview

Finding a safe, affordable, and available parking space is a common problem in busy cities, commercial areas, hospitals, colleges, tourist locations, and residential neighborhoods. Drivers often spend unnecessary time searching for parking because information about nearby parking availability is fragmented or unavailable. At the same time, many private property owners have unused parking spaces that remain vacant for several hours each day.

2. Core Problem Statement

There is no simple and intelligent platform that efficiently connects vehicle owners who need temporary parking with private space owners who have unused parking capacity, while also providing personalized parking recommendations, smart pricing assistance, and trust-oriented decision support.

3. Problems Faced by Drivers

- Difficulty locating nearby parking spaces in unfamiliar areas.
- Time and fuel wasted while manually searching for parking.
- Lack of clear information about price, amenities, vehicle compatibility, and availability.
- Uncertainty about parking safety and reliability.
- Generic search results that do not consider preferences such as CCTV, affordability, or vehicle type.

4. Problems Faced by Parking Space Owners

- Unused residential or commercial parking spaces generate no value.
- Owners lack a dedicated system to list, manage, and control temporary parking availability.
- Choosing a suitable hourly price can be difficult without market context.
- Manual booking coordination is inefficient and difficult to scale.

5. Proposed Solution – ParkShare

ParkShare is a digital parking marketplace that allows private parking space owners to register their available spaces and enables drivers to discover suitable parking options. The platform combines a modern web application, FastAPI backend, PostgreSQL database, and IBM watsonx.ai to support intelligent parking decisions.

The system uses IBM Granite through watsonx.ai to process structured parking and user preference data. AI-assisted features include personalized parking recommendations, smart price suggestions, and trust-related insights. Backend validation checks AI output against real application data before it is returned to the user.

6. How ParkShare Solves the Problem

Existing Problem	ParkShare Solution
Drivers cannot easily find suitable parking	Search and discover listed private parking spaces
Parking results are generic	AI-assisted recommendations based on user preferences
Owners do not know how to price spaces	Smart price suggestion support
Unused private parking remains vacant	Owners can list and monetize available space
Users lack confidence in parking options	Trust-oriented data and validated platform records

7. Role of Artificial Intelligence

Traditional parking applications mainly display listings or map locations. ParkShare extends this approach by using AI as a decision-support layer. IBM watsonx.ai and the Granite model analyze parking context and structured user preferences to generate more relevant recommendations. For example, a user requesting secure and affordable car parking with CCTV can receive a ranked recommendation based on matching parking data.

AI does not directly override core application data. Parking IDs, prices, and other critical values are validated by the backend. This reduces the risk of invented or invalid AI responses.

8. Expected Impact

- Reduce the time drivers spend searching for parking.
- Improve utilization of existing private parking infrastructure.
- Create an additional income opportunity for parking space owners.
- Provide more personalized and data-driven parking choices.
- Demonstrate a practical real-world use of IBM watsonx.ai and Granite models.

9. Conclusion

ParkShare addresses two connected real-world problems: parking scarcity for drivers and underused private parking capacity for owners. By creating a shared parking marketplace and adding AI-assisted recommendation and pricing capabilities, the platform aims to make parking discovery more efficient, intelligent, and accessible.